

SkillClaw: Collective Skill Evolution for Multi-User LLM Agent Ecosystems

Closed-loop, evidence-driven skill updates via an agentic evolver

Agent Skills Remain Static After Deployment — Users Rediscover

Static skill hub

Skills are reusable—but remain unchanged after deployment

Users manually install from a centralized hub

Interaction solutions rarely persist beyond a session

Repeated failure/recovery patterns recur across users

Evolving ecosystem

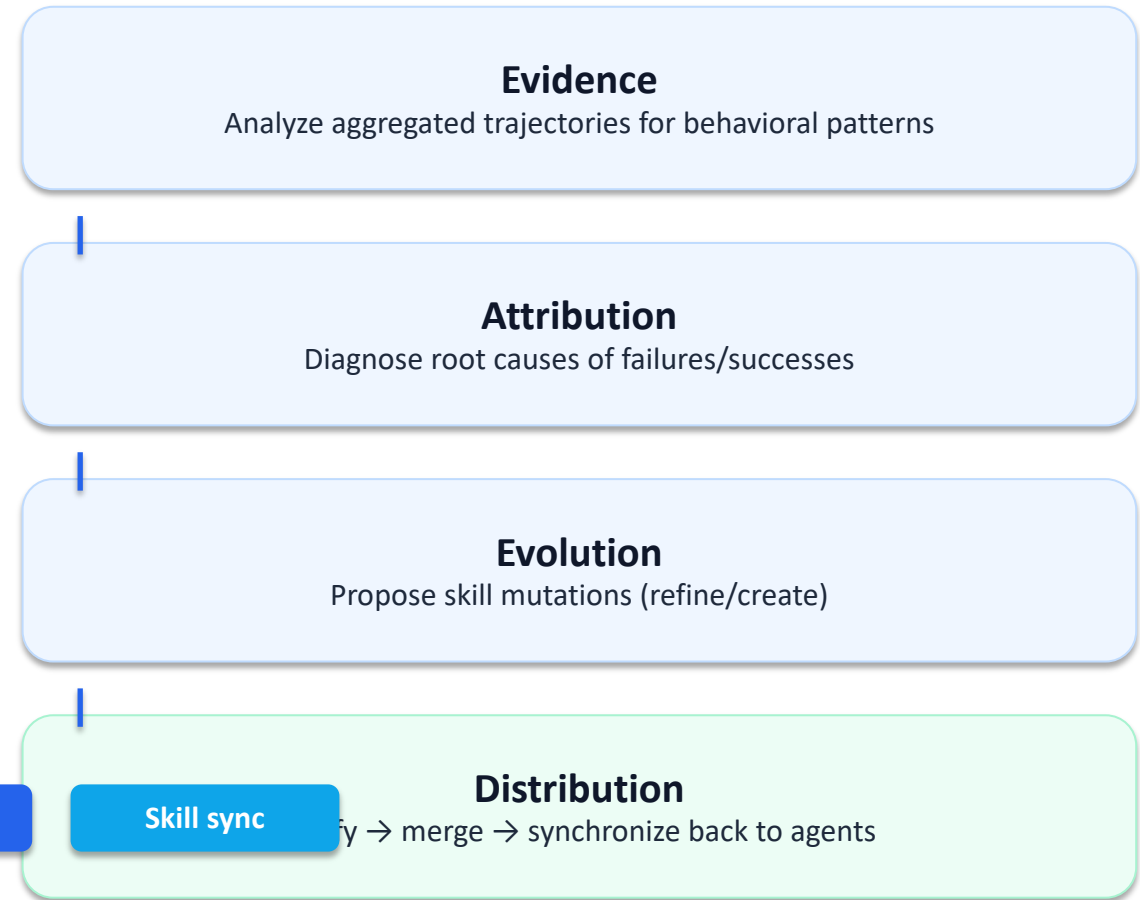
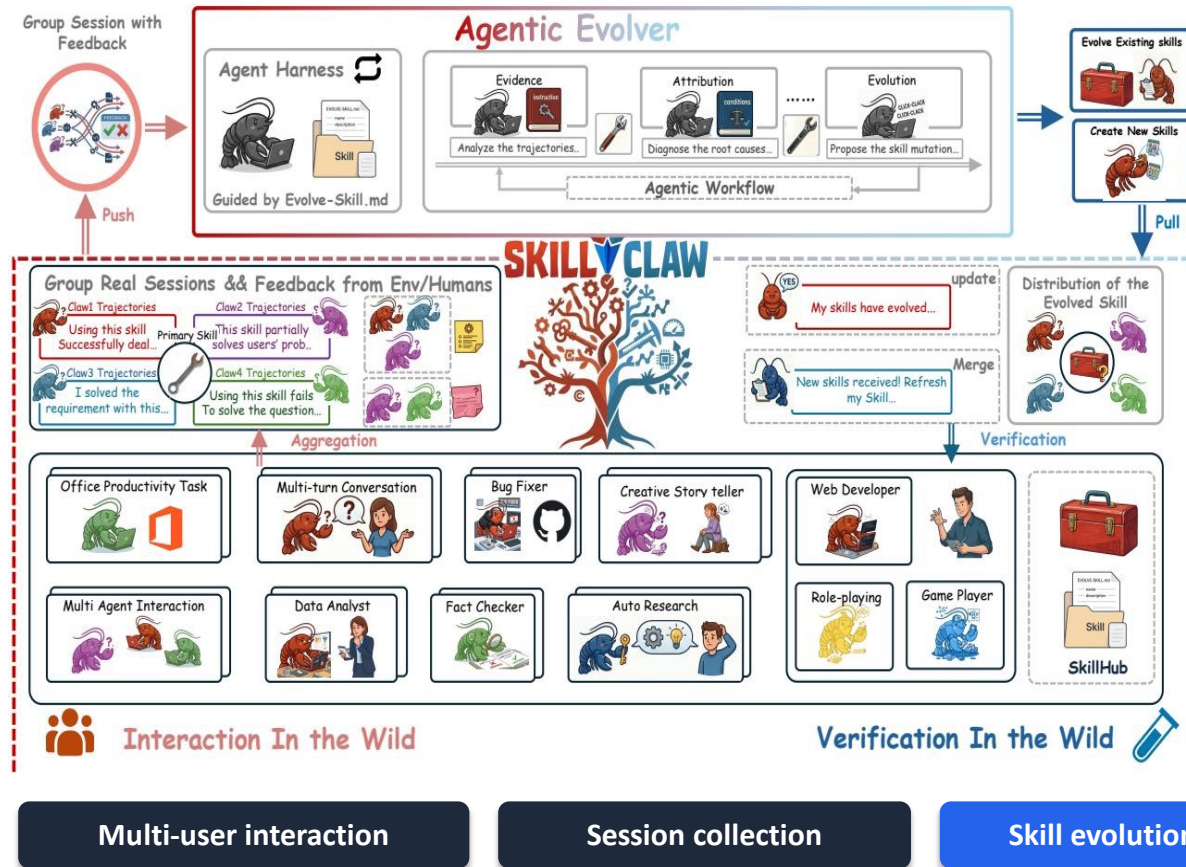
Aggregate evidence across users

Diagnose root causes; propose skill mutations

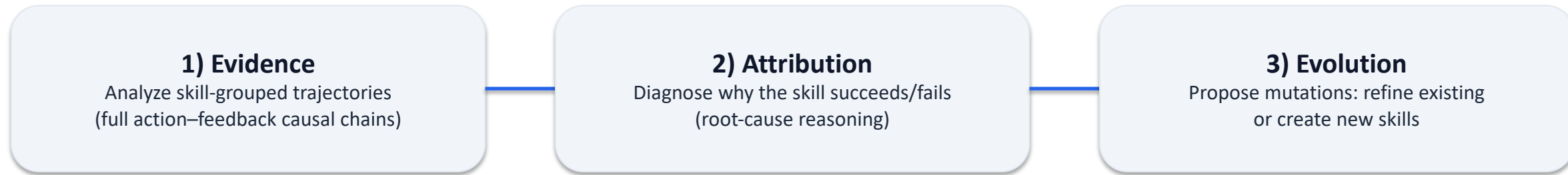
Verify updates to prevent regressions

Synchronize improved skills back to all agents

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill



Agentic Evolver: Evidence → Attribution → Evolution in Three Stages



Evolution is driven by an autonomous agent reasoning over evidence (not predefined update rules)

Updated skills are verified before merge into a shared repository

Goal: generalizable improvements, not instance-specific memory

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boun

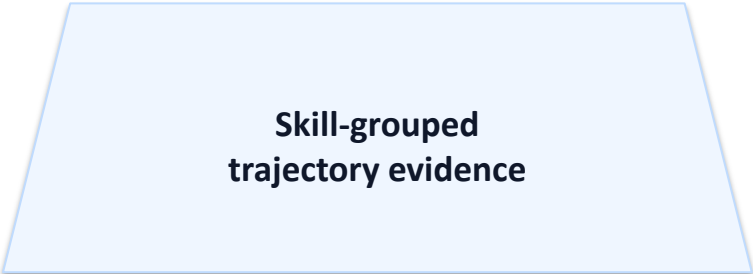
Independent agents produce structured session trajectories preserving action–feedback causal chains

Trajectories are aggregated across users and grouped by referenced skills to form shared evidence

Different contexts provide complementary signals: when a skill works vs. where it breaks

A single user rarely provides enough signal to separate generalizable fixes from idiosyncratic ones

Compatible with Claw-style systems: OpenClaw, CoPaw, IronClaw, PicoClaw, ZeroClaw, NanoClaw, NemoClaw



**Skill-grouped
trajectory evidence**

Case Study: Multimodal Creative Pipeline Skill — 6 Days of Evolution

Conservative verification: only 1 of 6 candidates accepted

Date	Outcome	Rationale (brief)
2026-01-01	Accepted	Accepted: validate /tmp_workspace inputs & environment setup
2026-01-02	Rejected	Rejected: multimodal input validation & output env init
2026-01-03	Rejected	Rejected: multimodal creative pipeline (extract from PDF/video/image)
2026-01-04	Rejected	Rejected: added image classification, visual generation, structured validation
2026-01-05	Rejected	Rejected: creative pipeline + per-file fail-fast validation
2026-01-06	Rejected	Rejected: extended PDF-to-poster / document-to-visual paths

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement

Iterative refinement: 3 of 6 days accepted before plateau

Date	Outcome	Rationale (brief)
2026-02-01	Accepted	Accepted: safe fallback instead of blocking on push failure
2026-02-02	Accepted	Accepted: unified patch/bundle filenames; reduced inconsistency
2026-02-03	Accepted	Accepted: auth-alternative paths & secrets audit; fixed subshell pitfalls
2026-02-04	Rejected	Rejected: same-pool retest (no better candidate admitted)
2026-02-05	Rejected	Rejected: treated push hang as auth failure; no improvement
2026-02-06	Rejected	Rejected: added identity config & filename consistency; no improvement

Key Takeaways: Three Properties That Distinguish SkillClaw from

Collective evolution: individual interactions contribute to a shared, continuously improving skill ecosystem

Fully automatic: evolution driven by runtime interaction without manual curation or explicit user intervention

Agentic evolution paradigm: open-ended reasoning over interaction evidence (not predefined update rules)

Implication: build agent systems that improve with use while preventing regressions via conservative verification